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APPLICATION EXAMPLE

Connecting WinCC Unified solutions to S7-1500R/H

WinCC Unified / S7-1500R/H

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1. Introduction

1.1. Use case scenario

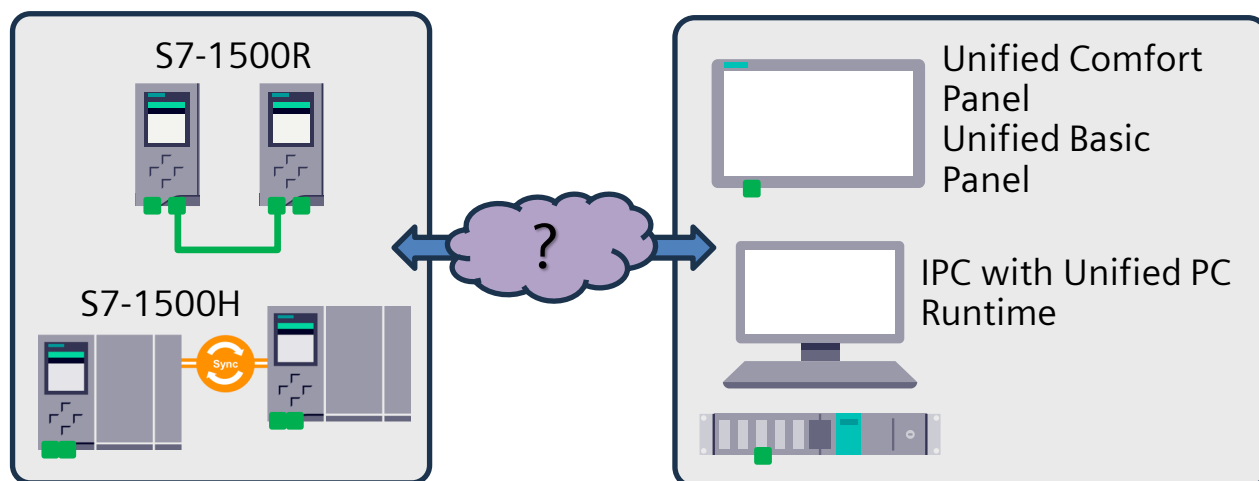
SIMATIC WinCC Unified devices enable machine-level process visualization and control. The same devices are used for both standard S7-1500 systems and S7-1500R/H redundant systems.

Through HMI communication, one or more WinCC Unified devices exchange data directly with the CPUs. Supported HMI devices include Unified Basic Panels, Unified Comfort Panels, and Industrial PCs running WinCC Unified Runtime.

In a S7-1500R/H redundant system, the automation process continues without interruption in the event of a CPU failure. Communication with HMI devices automatically switches to the remaining controller, ensuring continuous operation.

This application example demonstrates how to establish a redundant connection between SIMATIC WinCC Unified devices and a S7-1500R/H system.

Figure 1-1



1.2. Technical Solutions

This application example is based on two distinct technical approaches. Both methods are implemented to demonstrate different integration strategies and can be adapted to suit specific project requirements.

Device IP:

Each integrated interface of the CPU (X1, X2, X3) and virtual interface W1 of CP 1543-1 in a S7-1500R/H redundant system is assigned to a unique Device IP address. For instance, an S7-1500R system based on CPU S7-1515R includes two interfaces per R-CPU, resulting in four distinct Device IP addresses.

System IP:

In addition to the individual Device IP addresses, the S7-1500R/H system supports System IP addressing. This feature can be activated on each interface (X1, X2, X3 or W1).

HMI Connection Options:

SIMATIC HMI devices can be connected to the S7-1500R/H redundant system using one of the following methods:

- The HMI device communicates with the redundant system via **System-IP address**.
- The HMI device communicates directly with the individual R/H-CPU's via their respective **Device-IP addresses**.

1.3. System-IP vs. Device-IP

Each technical solution offers distinct characteristics and behaviors. To provide a clear overview, several key aspects are highlighted below.

NOTE

HMI communication via System-IP and Device-IP addresses are supported by all Unified solutions. That means this application example can be used with WinCC Unified Basic Panels, WinCC Unified Comfort Panels and WinCC Unified PC Runtime.

HMI Communication via System IP-Address:

- The HMI device communicates exclusively with the primary CPU using the System IP address.
- If the primary CPU fails, the backup CPU automatically assumes the role of the new primary and takes over the System IP address. Communication with the HMI device is re-established without manual intervention.
- No scripting is required. From the HMI device's perspective, the system behaves like a standard controller.
- However, if the primary CPU becomes unreachable (e.g., due to a switch failure), and no additional measures are implemented, HMI communication will be lost. If it is not acceptable by the user, the second solution via Devices IP-addresses should be preferred.

HMI Communication via Device IP-Address:

- The CPU role is irrelevant. The HMI device communicates with either the primary or the backup CPU using their respective Device IP addresses.
- In case of a CPU failure, a switchover mechanism ensures that communication is automatically redirected to the available CPU.
- This mechanism relies on scripting.
- Compared to the System IP approach, this method achieves faster switchover times.

NOTICE

Using X2 of an WinCC Unified Comfort Panel to establish a redundant physical connection is not supported.

1.4. Components used

This application example has been created with the following hard- and software components:

Table 1-1

Component	Article Number	Note
CPU 1513R-1 PN	6ES7513-1RM03-0AB0	FW used V4.0
CPU 1518HF-4 PN	6ES7518-4JT10-0AB0	FW used V4.0
SIMATIC HMI MTP1200 Unified Comfort	6AV2128-3MB06-0AXX	Image used V20.0.0.2
SIMATIC HMI MTP700 Unified Basic	6AV2123-3GB32-0AW0	Image used V20.0.0.2
SIMATIC IPC BX-39A	6AG4142-XXXXX-XXXX	Windows 10 IoT Enterprise LTSC 21H2
WinCC Unified PC (5k) RT V20	6AV2154-1FB02-4LA0	Image used V20.0.0.2
WinCC Unified PC (10k) ES V20	6AV2153-2FB02-4LA5	V20 Update 3
SIMATIC STEP7 Professional	6ES7822-1AE24-0YA5	V20 Update 3

1.5. Configuration Examples

The project includes four documented configuration options designed to cover a wide range of application scenarios. These configurations serve as templates and can be reused for similar implementations.

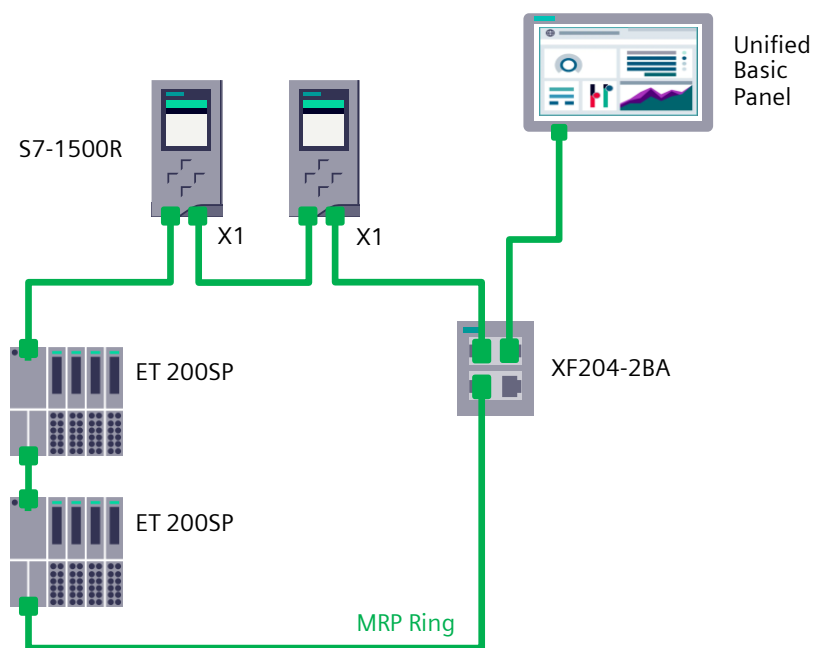
1.5.1. S7-1500R and Unified Basic Panel

In this section the Unified Basic Panel (UBP) is connected to the S7-1500R redundant system, via a SCALANCE switch, using the CPU interface X1.

The SCALANCE XF204-2BA and the ET 200SP stations act as a MRP Client.

NOTE The UBP does not support MRP configuration therefore cannot be part of the MRP ring.

Figure 1-2

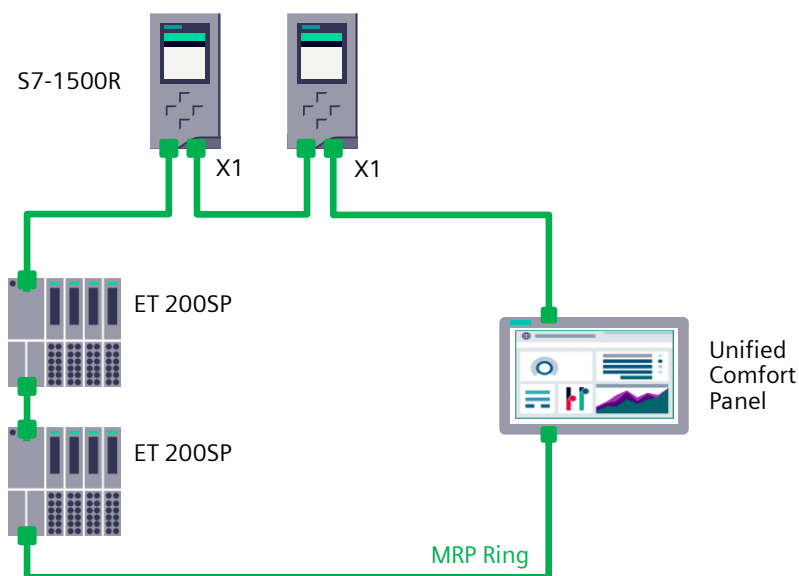


1.5.2. S7-1500R and Unified Comfort Panel

Through the MRP ring, the Unified Comfort Panel (UCP) is connected to the S7-1500R redundant system via the CPU interface X1.

The UCP as well as the ET 200SP stations act as a MRP Client.

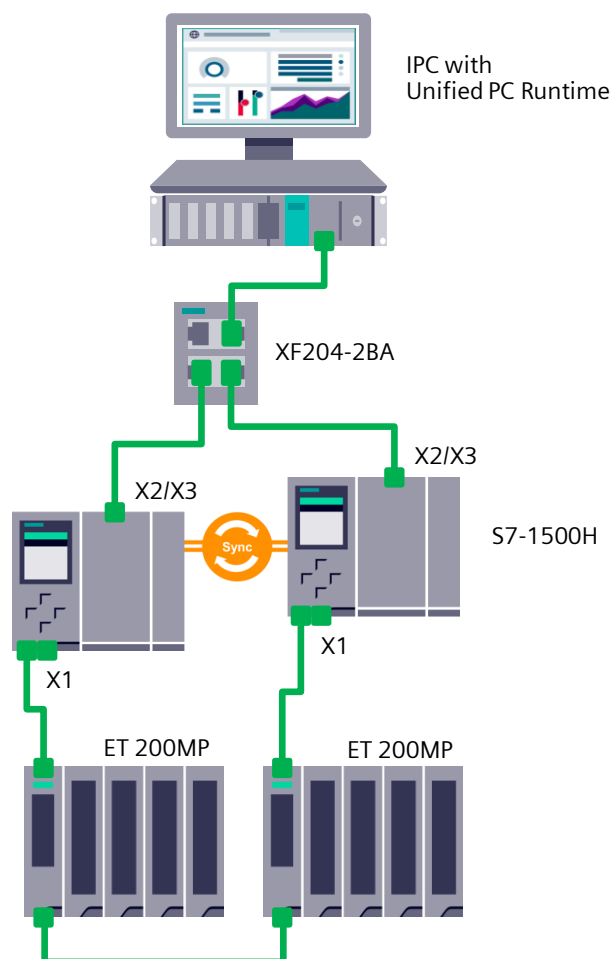
Figure 1-3



1.5.3. S7-1500H and IPC with Unified PC Runtime

The IPC, acting as an HMI device, is connected to the S7-1500H via a SCALANCE switch, using either the X2 or X3 interface of the H-CPU.

Figure 1-4



2. HMI Connection via System-IP address

2.1. Principle

The HMI Device communicates with the primary CPU via the system IP address in all operating modes: RUN-Redundant, RUN-Solo, and STOP. If the role of the primary CPU switches to the other CPU, the HMI Device automatically redirects its communication to the new primary CPU.

2.2. Configuration examples

NOTE

The configurations for this chapter can be found in the project "109781687_S7-1500RH_WinCC-Unified".

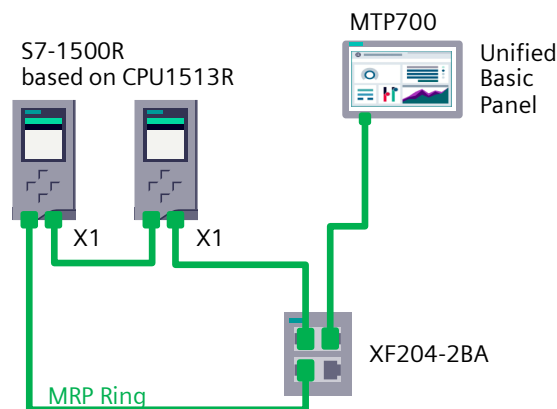
2.2.1. 01_S7-1500R – Unified Basic Panel

This section presents a compact configuration consisting of a **S7-1500R redundant system based on CPU1513R**, an **XF204-2BA switch**, and a **Unified Basic Panel MTP700**, positioned outside the MRP ring. The setup represents a simplified system layout that can be used as a reference for implementing redundant HMI communication.

While the structure is flexible, device compatibility must be considered. The HMI must be either a **Unified Panel** or a **PC-based Runtime (WinCC Unified PC RT)**. The controller must be a redundancy-capable CPU from the **S7-1500R/H series**.

The mechanism based on the **System IP address** remains fully supported, regardless of the hardware used, if the devices meet the technical requirements.

Figure 2-1



NOTE

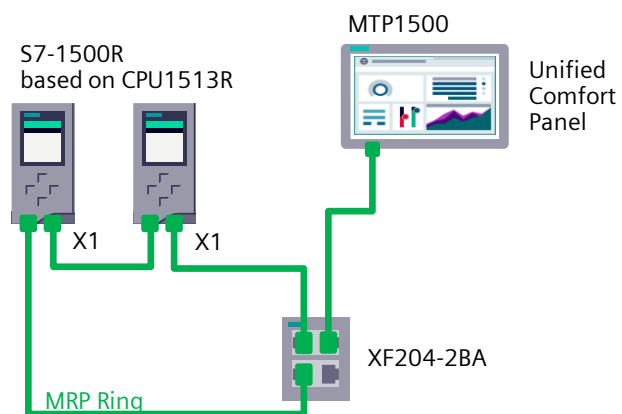
The configurations for this chapter can be found in the project "109781687_S7-1500RH_WinCC-Unified".

2.2.2. 02_S7-1500R – Unified Comfort Panel

Another configuration example included in the project features the same **S7-1500R redundant system based on CPU1513R and XF204-2BA switch** but uses a **Unified Comfort Panel MTP1500** instead of a Basic Panel. The panel is positioned outside the MRP ring, like the previous setup.

This configuration is provided to simplify the initial setup with a Comfort Panel and includes a sample project with an adapted screen resolution. Functionally, both configurations are identical and demonstrate the same communication behavior.

Figure 2-2

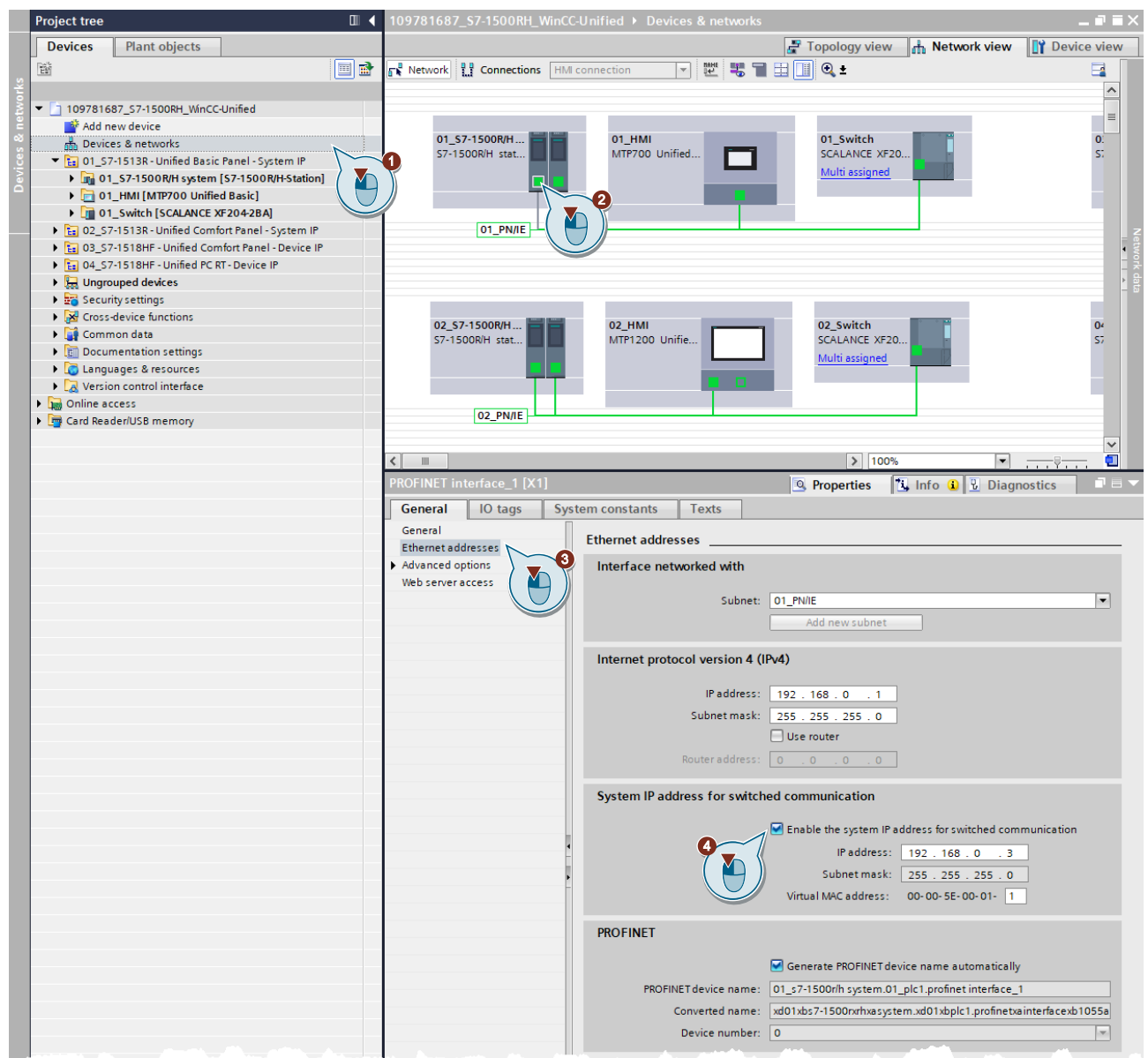


2.3. Enabling the System-IP address

To enable the System-IP address:

1. Use the project tree to open the device configuration “Devices & networks” (1).
2. Select the PLC “01_S7-1500 R/H system” / “02_S7-1500 R/H system” (2).
3. In the properties, open the menu “Ethernet address” (3).
4. Enable the option “Enable the system IP address for switches communication” (4).
The HMI Device will use this IP address to communicate with the S7-1500R/H redundant system.
Make sure that the System-IP address is in the same IP range as the Device-IP addresses.

Figure 2-3

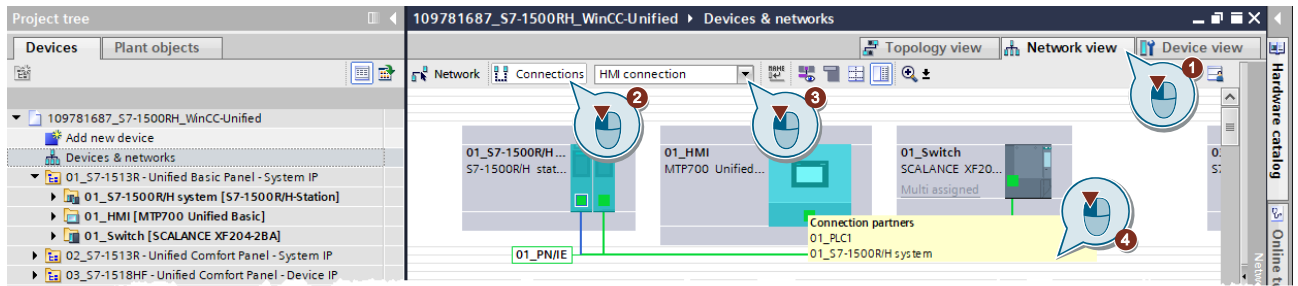


2.4. HMI connection via the System-IP address

To create an HMI connection via the system-IP address:

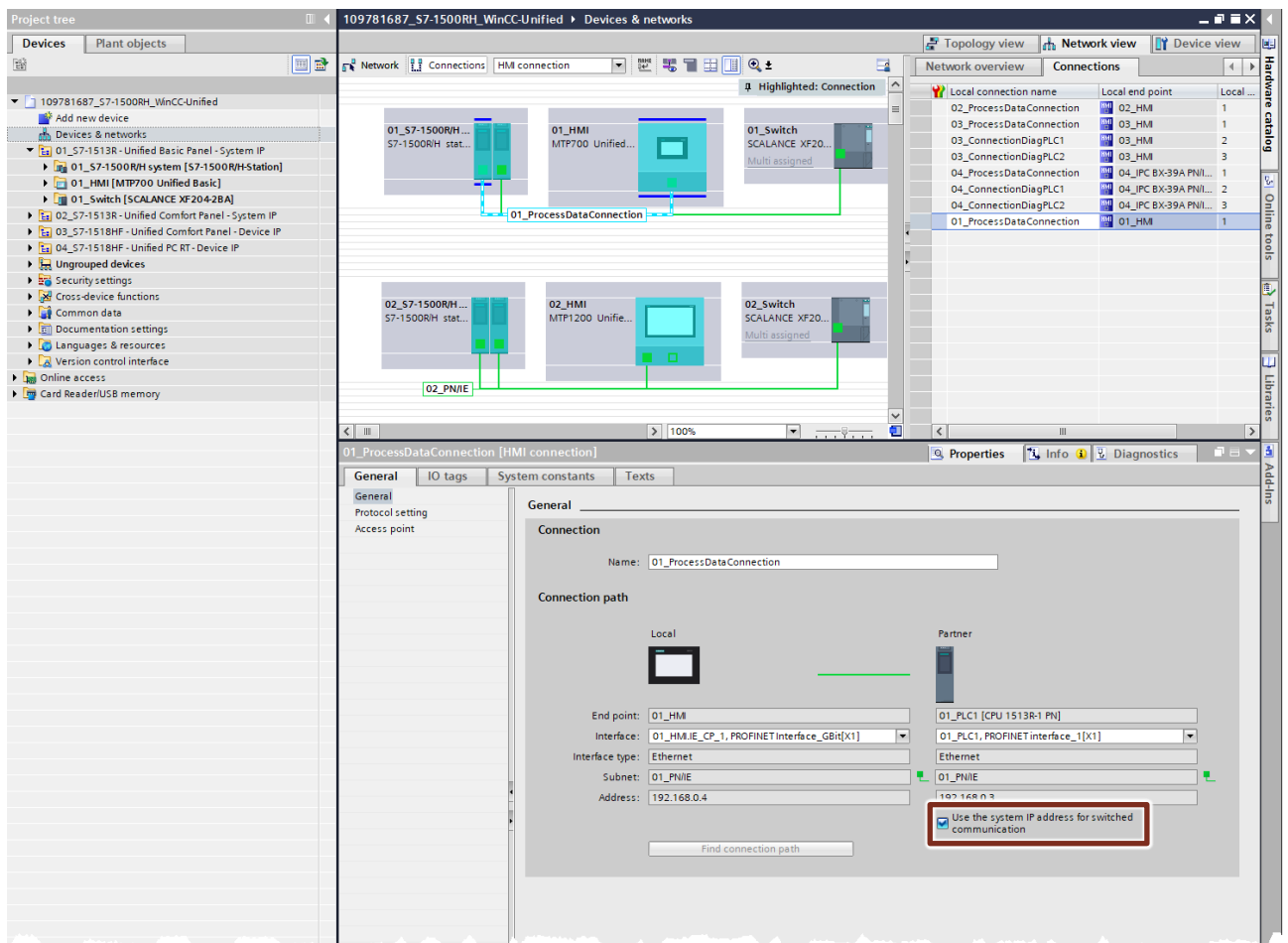
1. In the Network view (1), select the "Connections" tab (2), select HMI Connection (3).
2. Left click to select the "X1" interface of the first PLC, then hold down the left mouse button and drag a connection to the Ethernet interface of the HMI device.
3. A selection window opens. Here, select "01_S7-1500R/H-System" (4).

Figure 2-4



The settings in the configuration are now complete. The figure below shows an overview of the configured HMI connection.

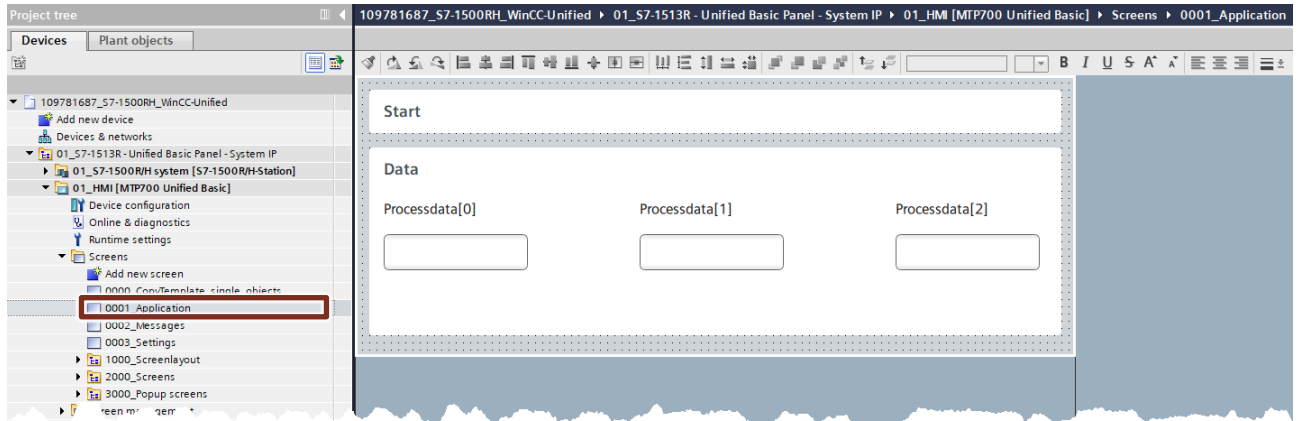
Figure 2-5



2.5. HMI-Screen for monitoring

The monitor screen can be edited within screen "0001_Application". Process values are displayed in three dedicated I/O fields. No additional configuration is required to enable switchover functionality.

Figure 2-6



3. HMI Connection via Device-IP addresses

3.1. Principle

The Device IP approach differs significantly from the System IP method due to its higher configuration effort. The required variables and scripts are described in section [Creating HMI-variables](#) and [Creation of the scheduled tasks](#).

In this setup, each PLC is assigned a dedicated HMI connection used to transmit a clock memory byte. These bytes serve as cyclic triggers to execute individual task scheduler routines. Each routine attempts to increase an internal counter on the opposite PLC while resetting its own counter to zero.

If one of the PLCs becomes unavailable, its corresponding task is no longer executed. As a result, the counter is not reset and continues to increase. Once a defined threshold is exceeded (internal tag: changePlcCounter), the system identifies the PLC as unreachable and initiates a ChangeConnection to switch communication to the available PLC.

Therefore, three HMI connections are required:

- **ConnectionDiagPLC1**: used to monitor the connection quality of **CPU1**,
- **ConnectionDiagPLC2**: used to monitor the connection quality of **CPU2**,
- **ProcessDataConnection**: dynamic connection used for accessing **process data**.

Depending on the connection quality, the HMI connection "**ProcessDataConnection**" will receive the parameters of the connection PLC1 or PLC2. This switchover connection is performed through **Scheduled Tasks** in the HMI project.

A 1st scheduled task "**connectionPLC1**" is triggered by the clock bit information ("M0.5" / 500ms) over the connection "**connectionPLC1**". In this task a counter "**plc1counter**" is set to "0" and a second "**plc2counter**" is incremented.

A 2nd scheduled task "**connectionPLC2**" is triggered by the clock bit information ("M0.5" / 500ms) over the connection "**connectionPLC2**". In this task a counter "**plc2counter**" is set to "0" and a second "**plc1counter**" is incremented.

A 3rd scheduled task "**connectionLost**" is triggered every 10 seconds. In this task, the counter, "**plc1counter**" and "**plc2counter**" is incremented.

The variable changePlcCounter holds a static value that can be adjusted. It is not recommended to set the start value below 3, as two missed cycles may result in a switchover that is too fast. If necessary, the start value can be increased. The default value in the demo project is 3.

Table 3-1

Case	Counter	State	Comments
PLC1 is connected PLC2 is connected	plc1counter < changePlcCounter plc2counter < changePlcCounter	No connection failure	Communication via PLC1
PLC1 is disconnected PLC2 is connected	plc1counter > changePlcCounter plc2counter < changePlcCounter	Connection failure with PLC1 → switchover connection PLC2	Communication via PLC2
PLC1 is connected PLC2 is disconnected	plc1counter < changePlcCounter plc2counter > changePlcCounter	Connection failure with PLC2 → switchover connection PLC1	Communication via PLC1
PLC1 is disconnected PLC2 is disconnected	plc1counter > changePlcCounter plc2counter > changePlcCounter	No connection available	No communication

A 4th scheduled task "**connectionHMI**" is triggered by the clock bit information ("M0.5" / 500ms) over the connection "**ProcessDataConnection**". This task is used to reset the status after a lost connection.

3.2. Configuration examples

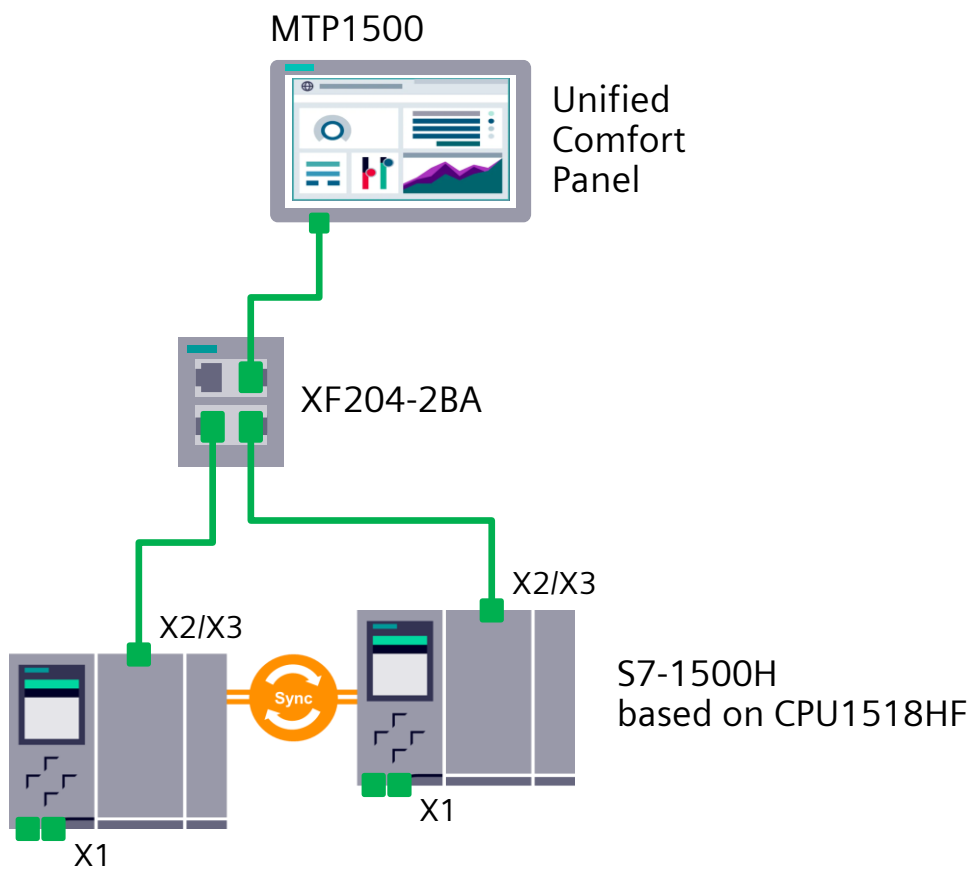
NOTE

The configurations for this chapter can be found in the project “109781687_S7-1500RH_WinCC-Unified”.

3.2.1. 03_S7-1500H – Unified Comfort Panel

This configuration includes a Unified Comfort Panel MTP1500, a **S7-1500H redundant system based on CPU1518HF**, and an XF204-2BA switch. The setup follows a similar structure to the previous examples, with the panel positioned outside the MRP ring.

Figure 3-1



NOTE

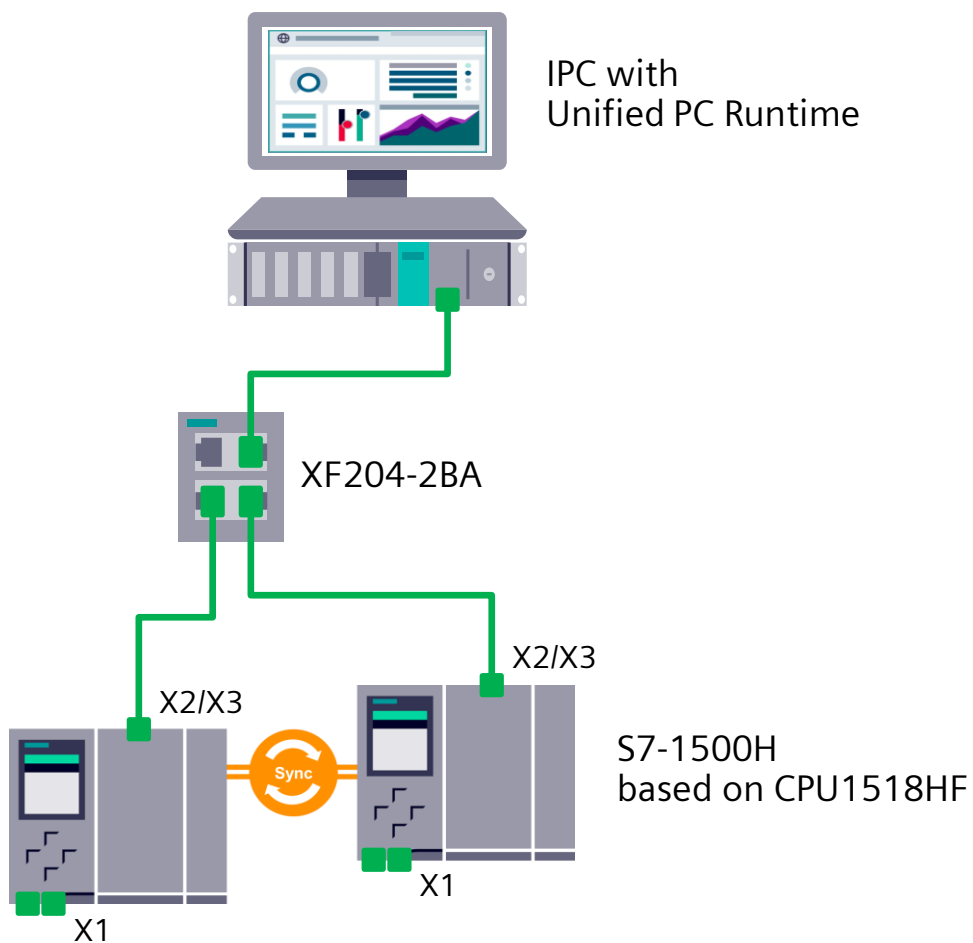
The configurations for this chapter can be found in the project “109781687_S7-1500RH_WinCC-Unified”.

3.2.2. 04_S7-1500H – Unified PC RT

This configuration includes an **IPC BX-39A** running **WinCC Unified PC Runtime**, an **S7-1500H redundant system based on CPU1518HF**, and an **XF204-2BA** switch. The IPC is positioned outside the MRP ring, like the previous examples.

This setup is provided as a template for projects using an Industrial PC instead of a Unified Panel. Apart from the HMI device, the configuration remains consistent with the previous examples and demonstrates how an IPC-based solution can be integrated into a redundant system.

Figure 3-2

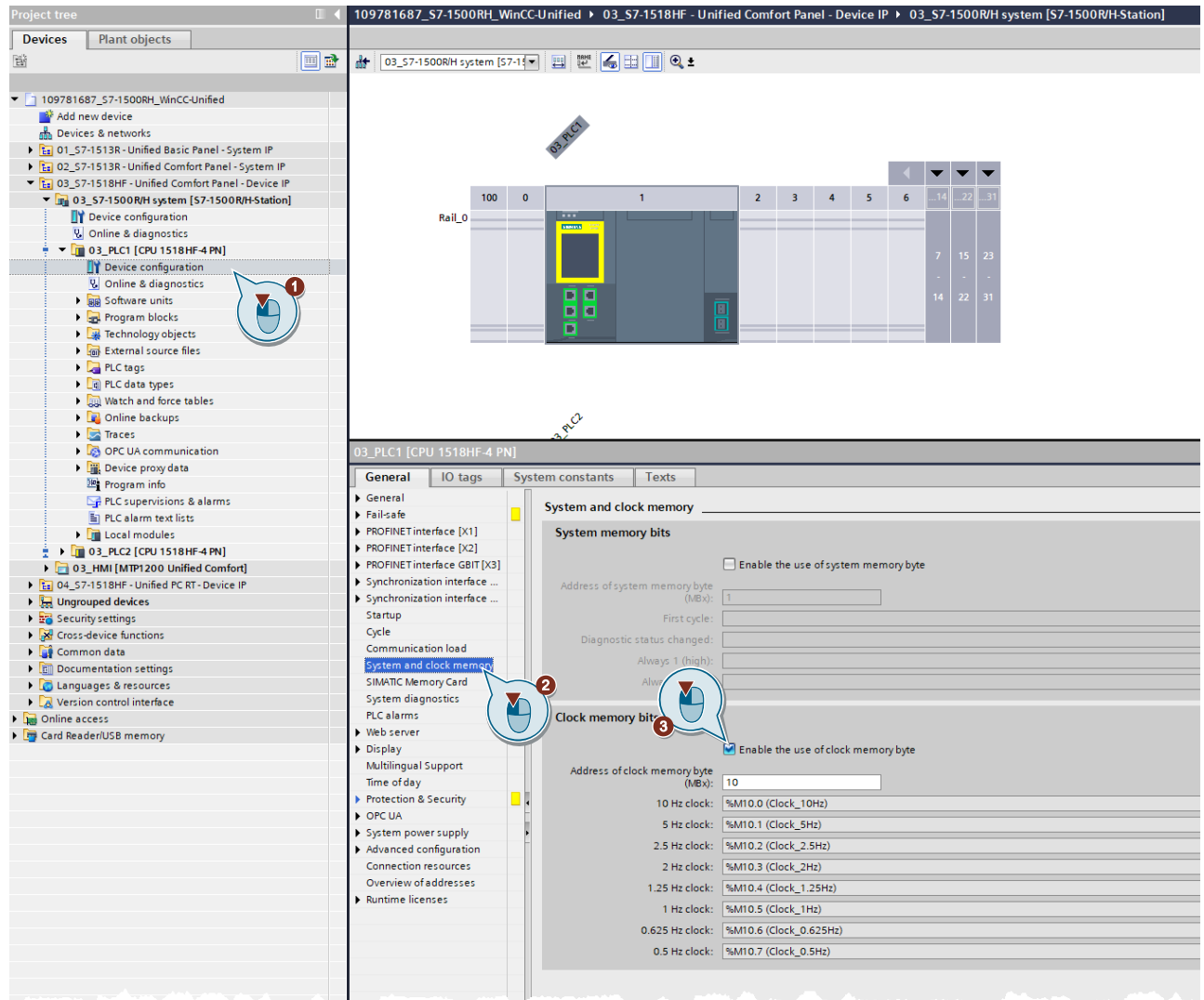


3.3. Enabling of the clock byte variable

To enable the memory clock:

1. Use the project tree to open the device configuration for "03_PLC1/04_PLC1" (1).
2. In the properties, open the menu "System and clock memory" (2).
3. Activate "Enable the use of the clock memory byte" (3).

Figure 3-3



3.4. The “three” HMI connections over the Device-IP addresses

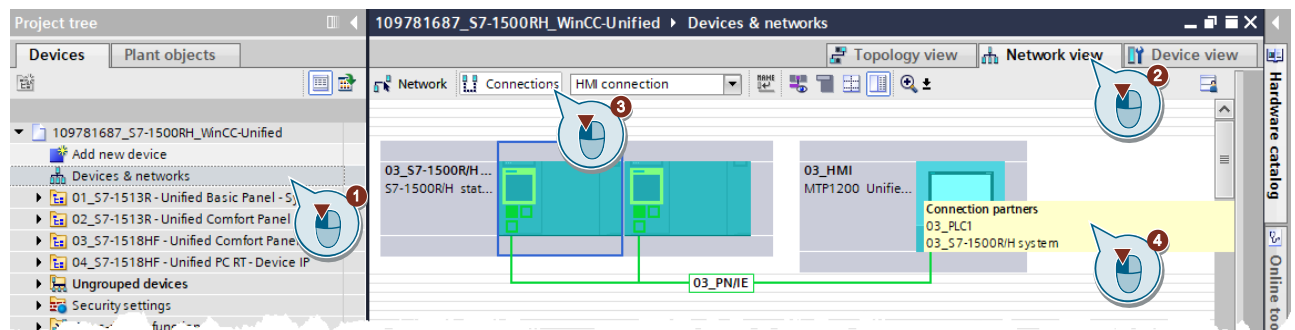
Each of these configurations requires three separate HMI connections to reliably determine the availability of PLC1 and PLC2, and to access process data.

To accurately evaluate the connection status of PLC1 and PLC2, each memory clock bit must be linked to a dedicated HMI connection. Without this setup, connection changes cannot be reliably detected.

To create an HMI connection “03_ConnectionDiagPLC1”:

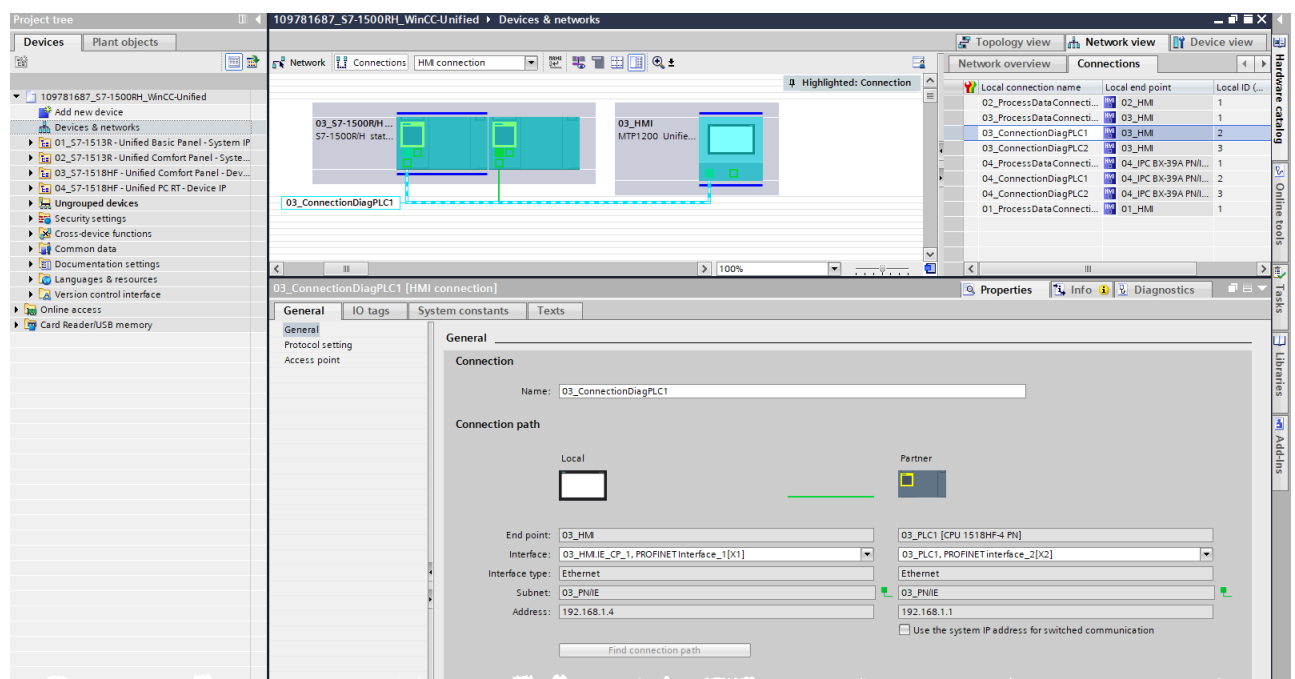
1. Use the project tree to open the “Devices & networks” (1).
2. In the Network view (2), select the “Connections” tab (3) and select HMI Connection.
3. Drag and drop the connection from the left PLC to the Unified Comfort Panel.
4. A selection window opens. Here, select “03_PL1” (4).

Figure 3-4



The settings in the configuration “03_ConnectionDiagPLC1” are now complete. The figure below shows an overview of the configured HMI connection.

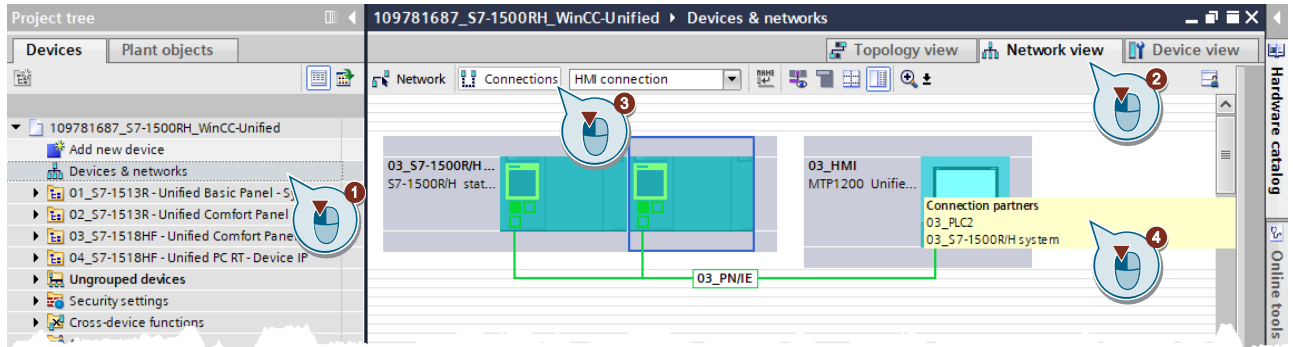
Figure 3-5



To create an HMI connection "03_ConnectionDiagPLC2 / 04_ConnectionDiagPLC2":

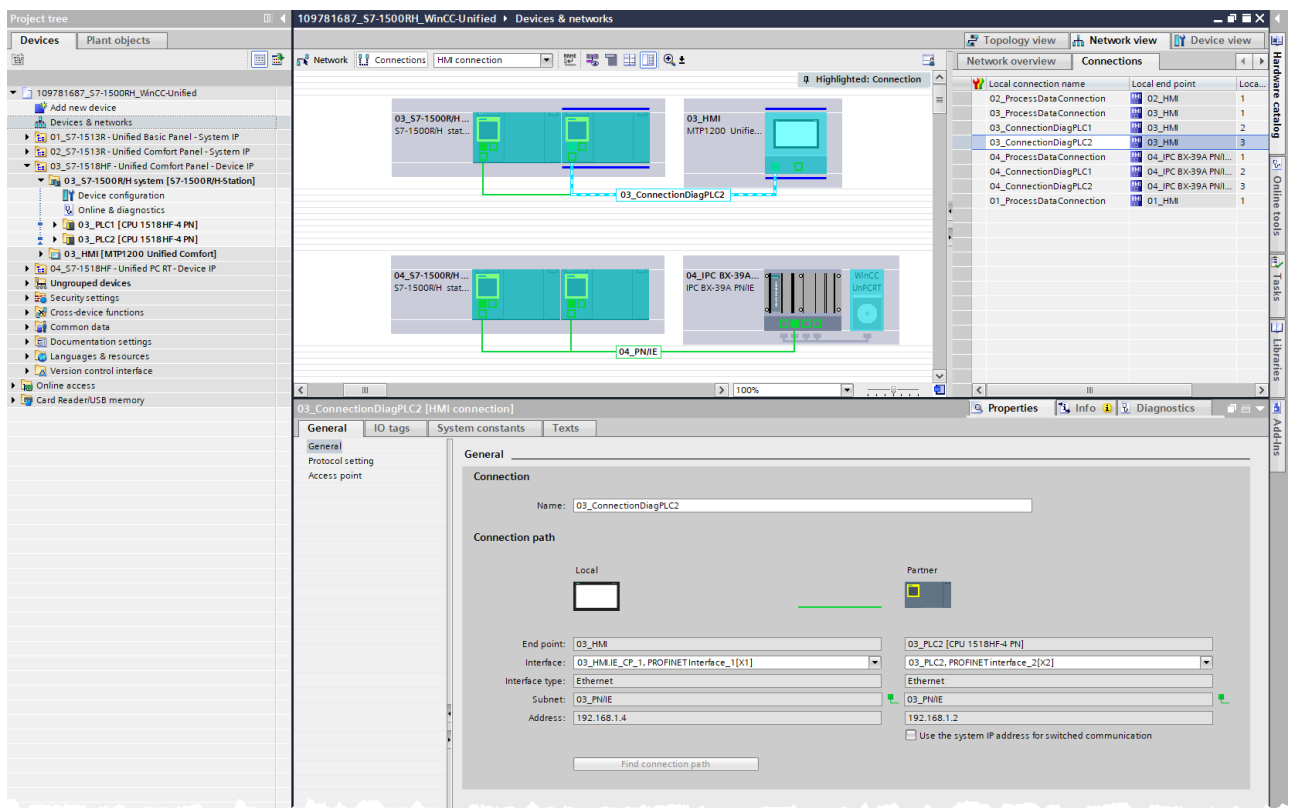
1. Use the project tree to open the "Devices & networks" (1).
2. In the Network view (2), select the "Connections" tab (3) and select "HMI Connection".
3. Drag and drop the connection from the right PLC to the Unified Comfort Panel.
4. A selection window opens. Here, select "03_PLC2" (4).

Figure 3-6



The settings in the configuration "03_ConnectionDiagPLC2" are now complete. The figure below shows an overview of the configured HMI connection.

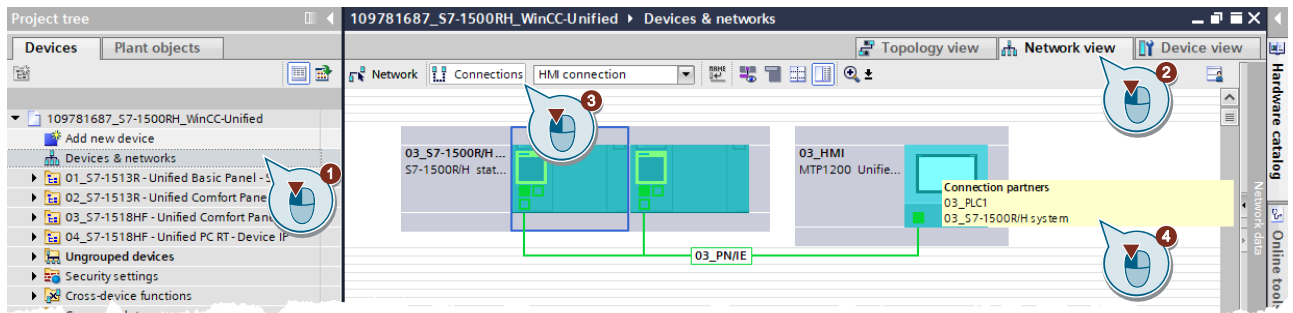
Figure 3-7



To create an HMI connection "03_ConnectionDiagPLC2":

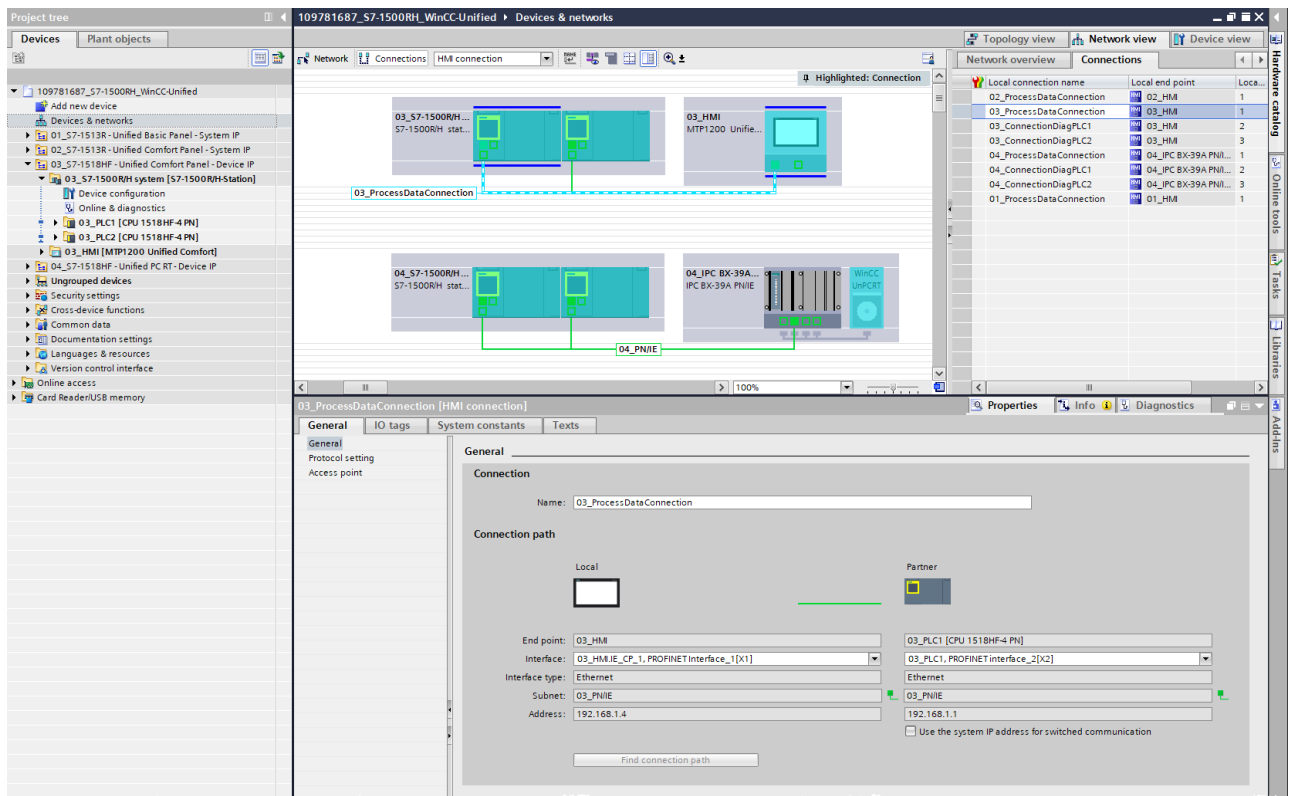
1. Use the project tree to open the "Devices & networks" (1).
2. In the Network view (2), select the "Connections" tab (3) and select "HMI Connection".
3. Drag and drop the connection from the left PLC to the Unified Comfort Panel.
4. A selection window opens. Here, select "03_S7-1500R/H system" (4).

Figure 3-8



The settings in the configuration "03_ProcessDataConnection / 04_ProcessDataConnection" are now complete. The figure below shows an overview of the configured HMI connection.

Figure 3-9



3.5. Creating HMI-variables

3.5.1. For the configuration “01_Config”

To apply this method in other projects, all necessary adjustments can be made in a single location. The scripts reference three internal variables, and modifying the initial value of these variables is sufficient to propagate the changes across all scripts.

Figure 3-10

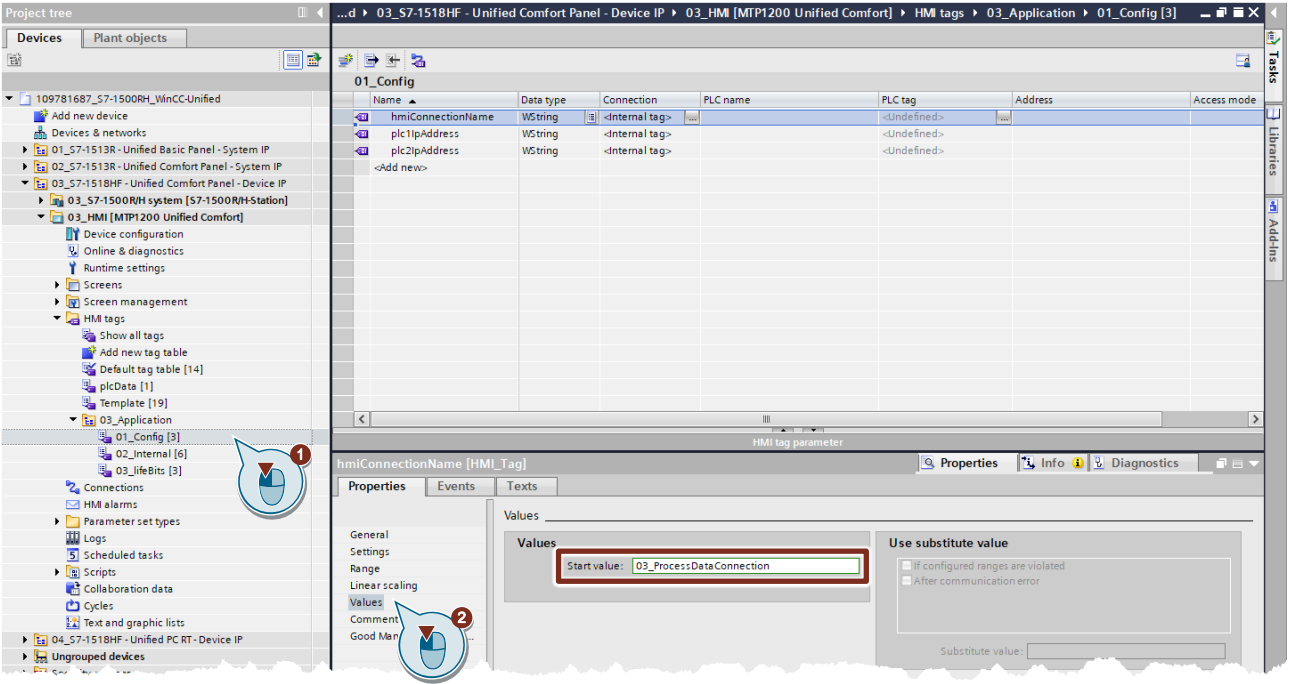


Table 3-2

Tag	Data type	Connection	Comment	Start value
hmiConnectionName	WString	Internal	Stores the name of the HMI Connection	03_ProcessDataConnection
plc1IpAdress	WString	internal	Stores the IP-Address for PLC1	192.168.1.1
plc2IpAdress	WString	internal	Stores the IP-Address for PLC2	192.168.1.2

To customize the demo project, the starting value of the following three variables must be adjusted.

1. Open the Tag editor via “Project tree > 03_HMI > HMI tags > Application > 01_Config
2. In the properties, open the menu “Values”
3. Specify the required start value to apply the configuration consistently across all relevant scripts

NOTICE The same steps can be taken to modify the IP addresses of the PLCs.

Figure 3-11

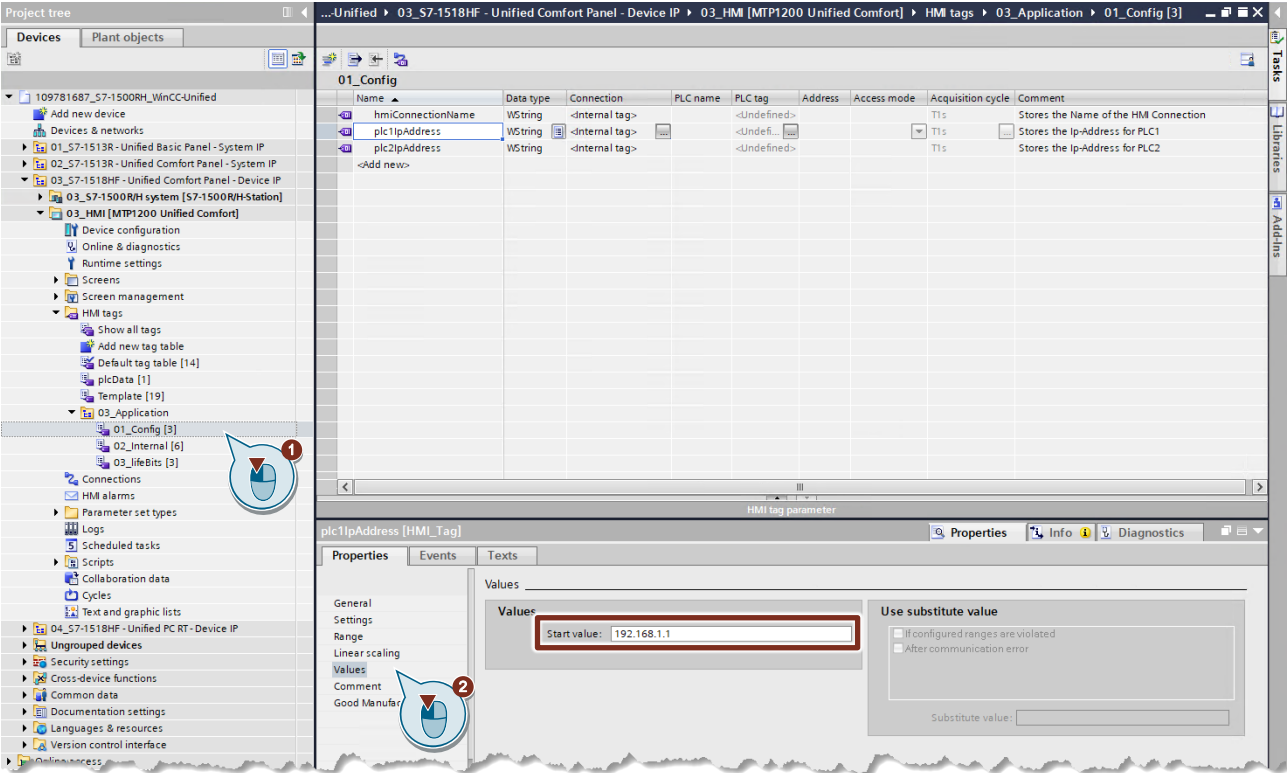
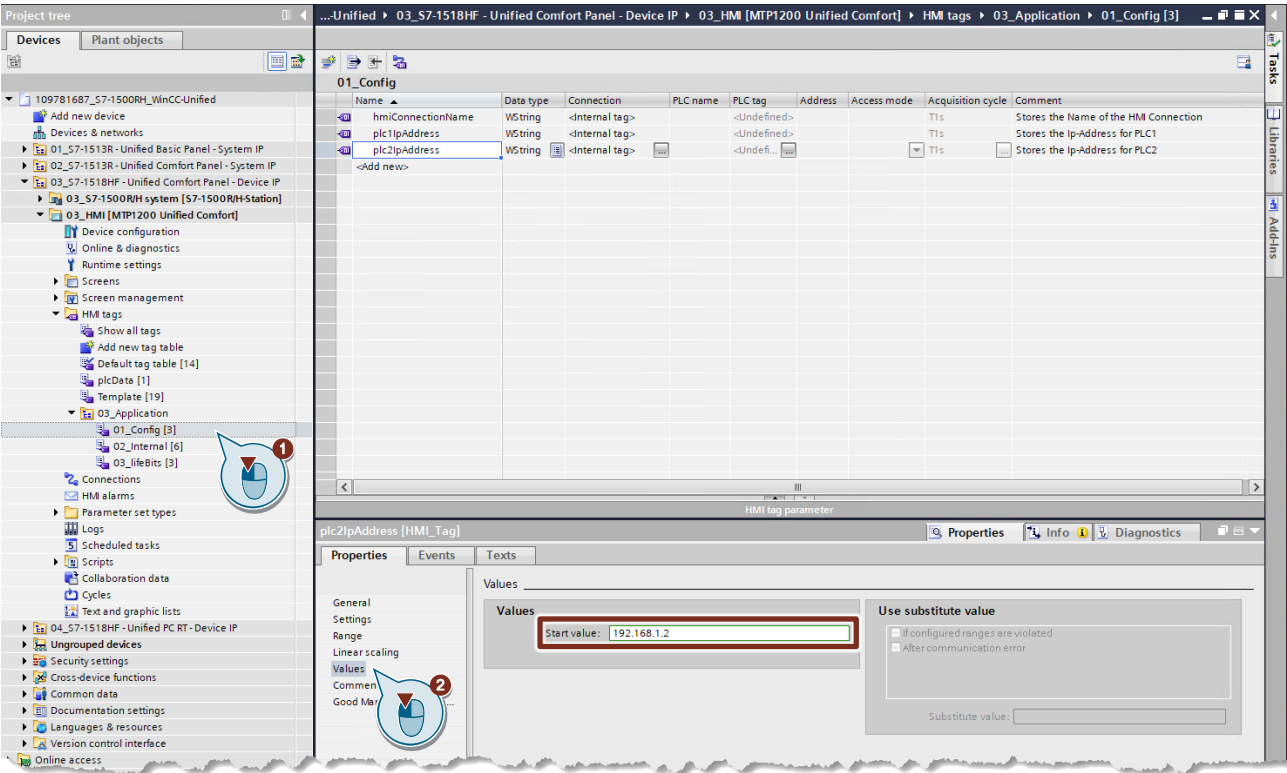


Figure 3-12



3.5.2. For the configuration “02_Internal”

To ensure proper script functionality, the following internal variables must be present in the project. No modifications are necessary; the variables simply need to exist within the project structure.

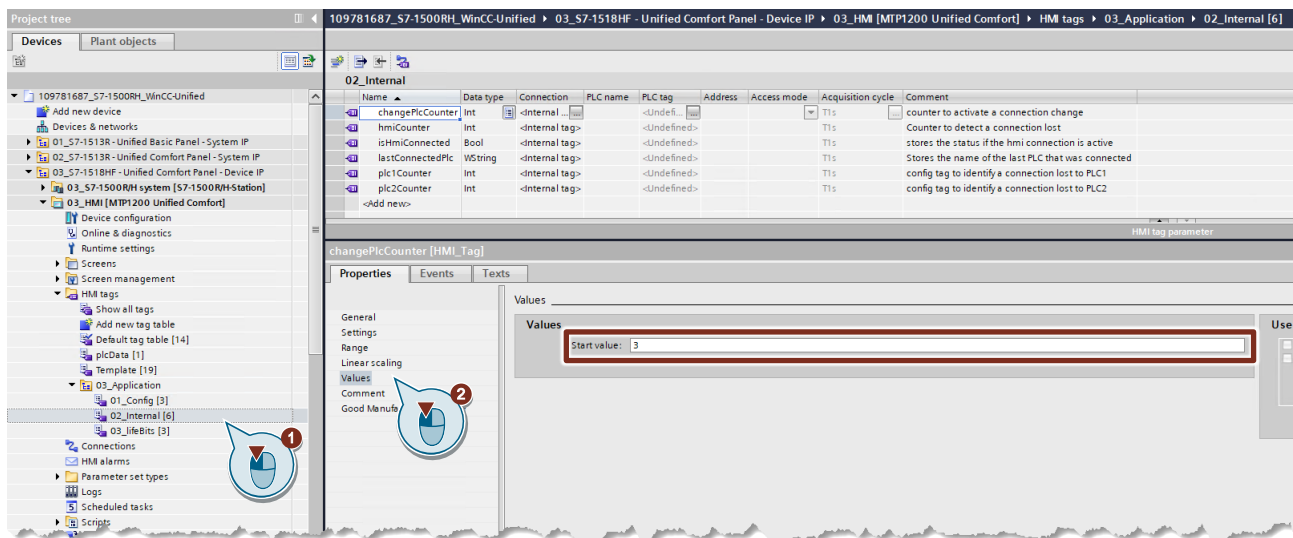
Table 3-3

Tag	Data type	Connection	Comment
isHmiConnected	Bool	Internal	Stores the status if the connection is active
plc1Counter	Integer	Internal	Config tag to identify a connection lost to PLC1
plc2Counter	Integer	Internal	Config tag to identify a connection lost to PLC2
changePlcCounter	Integer	Internal	Counter to activate a connection change
hmiCounter	Integer	Internal	Counter to detect a connection lost
lastConnectedPlc	WString	Internal	Stores the name of the last PLC that was connected

If the connection switch occurs too fast, the sensitivity can be reduced by increasing the start value of the “changePlcCounter” variable, for instance, from 3 to 6.

1. Open the Tag editor via “Project tree > 03_HMI > HMI tags > Application > 02_Internal
2. In the properties, open the menu “Values”
3. Specify the required start value to apply the configuration consistently across all relevant scripts

Figure 3-13



3.5.3. For the configuration “03_LifeBits”

Lifebits play a critical role in this solution, as they reflect the quality of each HMI connection. It is essential to ensure that the correct HMI connection is used for each lifebit to guarantee reliable status evaluation.

Table 3-4

Tag	Data type	Connection	Comment
isHmiConnectionAlive	Bool	03_ProcessDataConnection	Indicates whether the HMI connection is currently active (Lifebit)
isPlc1Alive	Bool	03_ConnectionDiagPLC1	Indicates whether PLC1 is currently active (Lifebit)
isPlc2Alive	Bool	03_ConnectionDiagPLC2	Indicates whether PLC2 is currently active (Lifebit)

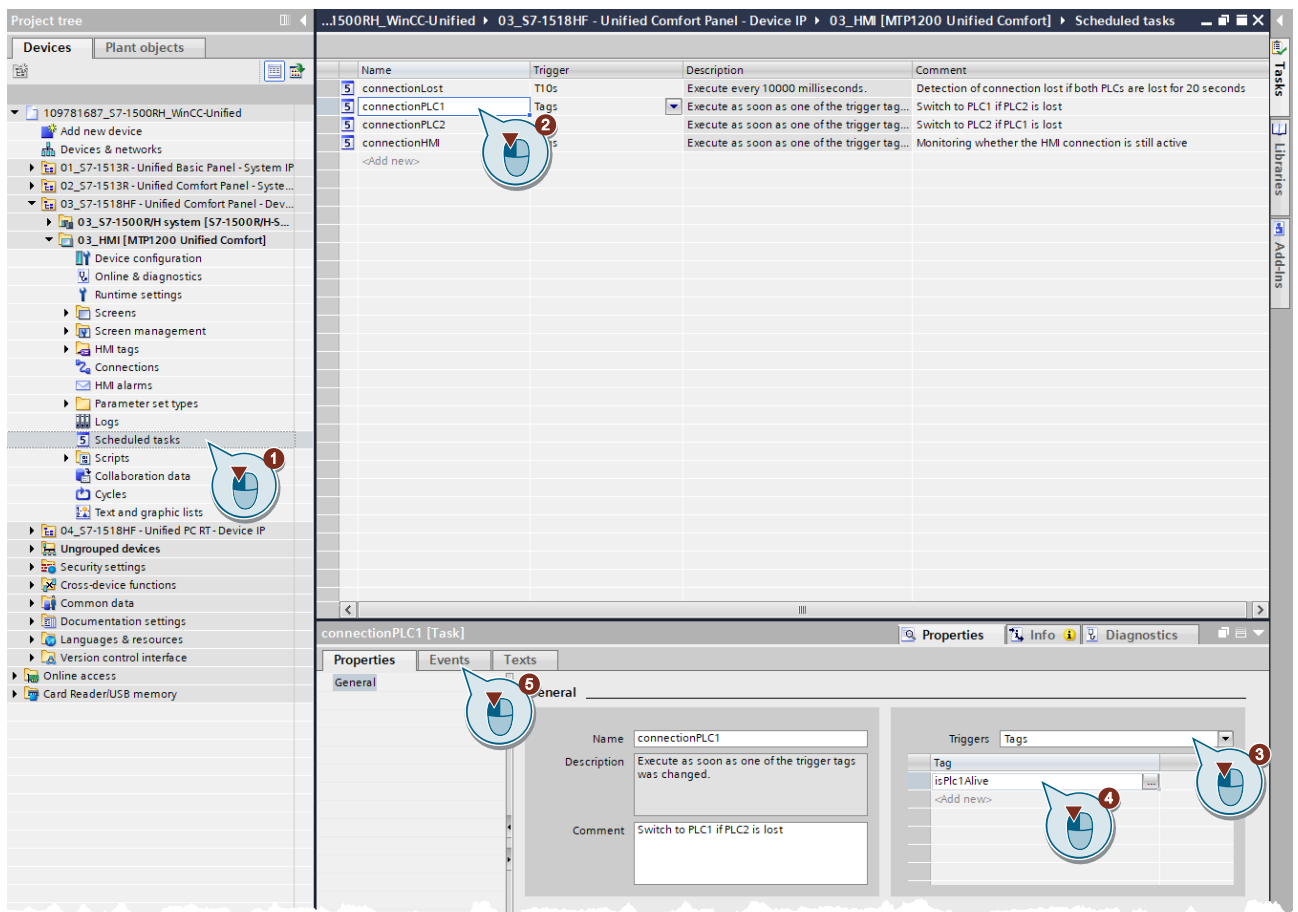
3.6. Creation of the scheduled tasks

3.6.1. Scheduled tasks - connectionPLC1

To edit the scheduled tasks:

1. Open the Scheduled Tasks section in the project tree
2. Select the relevant scheduled task
3. Use tags as triggers
4. Use the variable introduced in [For the configuration "01_Config"](#), which contains the memory clock bit from PLC1 and is linked to the corresponding HMI Connection
5. The code is located under Events

Figure 3-14



```

let ts = Tags.CreateTagSet([
    "plc1Counter",           // ts[0]
    "plc2Counter",           // ts[1]
    "lastConnectedPlc",      // ts[2]
    "plc1IpAddress",         // ts[3]
    "plc2IpAddress",         // ts[4]
    "hmiConnectionName",    // ts[5]
    "changePlcCounter"       // ts[6]
]);
ts.Read();

// Initialization
if (ts[2].Value === "")      // lastConnectedPlc
    HMIRuntime.Connections.SysFct.ChangeConnection(ts[5].Value, ts[3].Value);
                                // hmiConnectionName, plc1IpAddress
    ts[2].Value = "Connected with PLC1";
                                // lastConnectedPlc
}

// Reset of the connection lost detection for PLC1
ts[0].Value = 0;             // plc1Counter

// Connection lost detection for PLC2 - Incrementing the plc2Counter
if (ts[1].Value <= ts[6].Value) { // plc2Counter, changePlcCounter
    ts[1].Value = ts[1].Value + 1; // plc2Counter
}

// Switch to PLC1 if PLC2 is lost
if (ts[1].Value > ts[6].Value && ts[2].Value === "Connected with PLC2") {
    // plc2Counter, changePlcCounter,
    lastConnectedPlc
    HMIRuntime.Connections.SysFct.ChangeConnection(ts[5].Value, ts[3].Value);
                                // hmiConnectionName, plc1IpAddress
    ts[2].Value = "Connected with PLC1"; // lastConnectedPlc
    ts[1].Value = 0;                 // plc2Counter
}
ts.Write();

```

The system continuously reads various tags, including IP addresses, counters, and connection status indicators. If no active connection is detected, the HMI automatically switches to PLC1.

To confirm PLC1 is operational, its internal counter (plc1Counter) is reset to zero. Simultaneously, the counter for PLC2 (plc2Counter) is incremented - provided it remains below a defined threshold (changePlcCounter). This behavior indicates that PLC2 is not sending a valid lifebit.

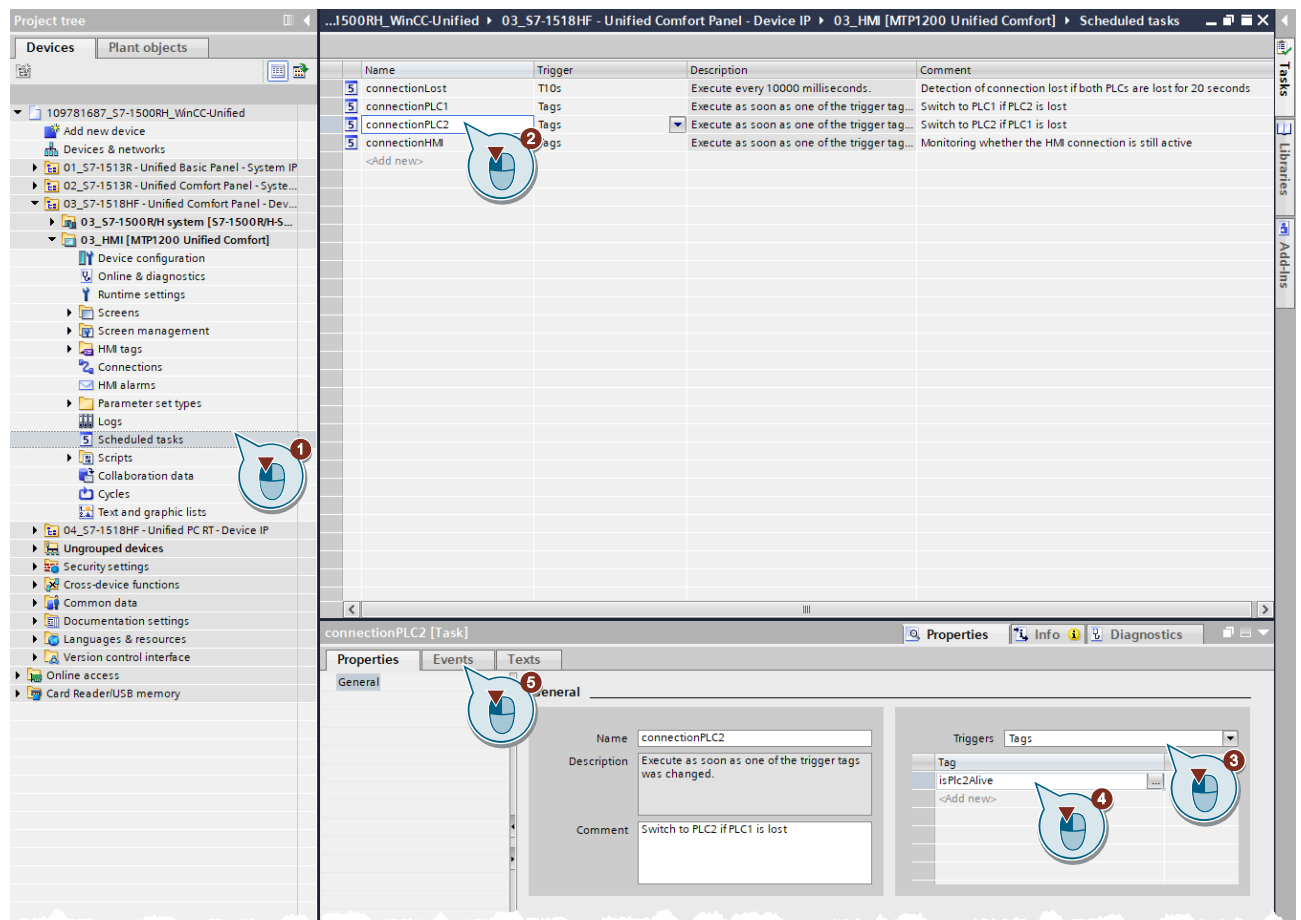
If PLC2 is marked as connected but its counter exceeds the threshold, the system interprets this as a loss of communication and initiates a switchover to PLC1.

3.6.2. Scheduled tasks - connectionPLC2

To edit the scheduled tasks:

1. Open the Scheduled Tasks section in the project tree
2. Select the relevant scheduled task
3. Use tags as triggers
4. Use the variable introduced in [For the configuration "01_Config"](#), which contains the memory clock bit from PLC2 and is linked to the corresponding HMI Connection
5. The code is located under Events

Figure 3-15



PLC2 resets its internal counter (plc2Counter) to zero to indicate that it is operational. At the same time, the counter for PLC1 (plc1Counter) is incremented if PLC1 fails to send a valid Lifebit.

If PLC1 is marked as connected but its counter exceeds a defined threshold, the system interprets this as a loss of communication and initiates a switchover to PLC2.

```
let ts = Tags.CreateTagSet([
    "plc1Counter",           // ts[0]
    "plc2Counter",           // ts[1]
    "lastConnectedPlc",      // ts[2]
    "plc1IpAddress",         // ts[3]
    "plc2IpAddress",         // ts[4]
    "hmiConnectionName",    // ts[5]
    "changePlcCounter"       // ts[6]
]);
ts.Read();

// Reset of the failure detection for PLC2
ts[1].Value = 0;           // plc2Counter

// Connection lost detection for PLC2 - Incrementing the plc2Counter
if (ts[0].Value <= ts[6].Value) {           // plc1Counter, changePlcCounter
    ts[0].Value = ts[0].Value + 1;         // plc1Counter
}

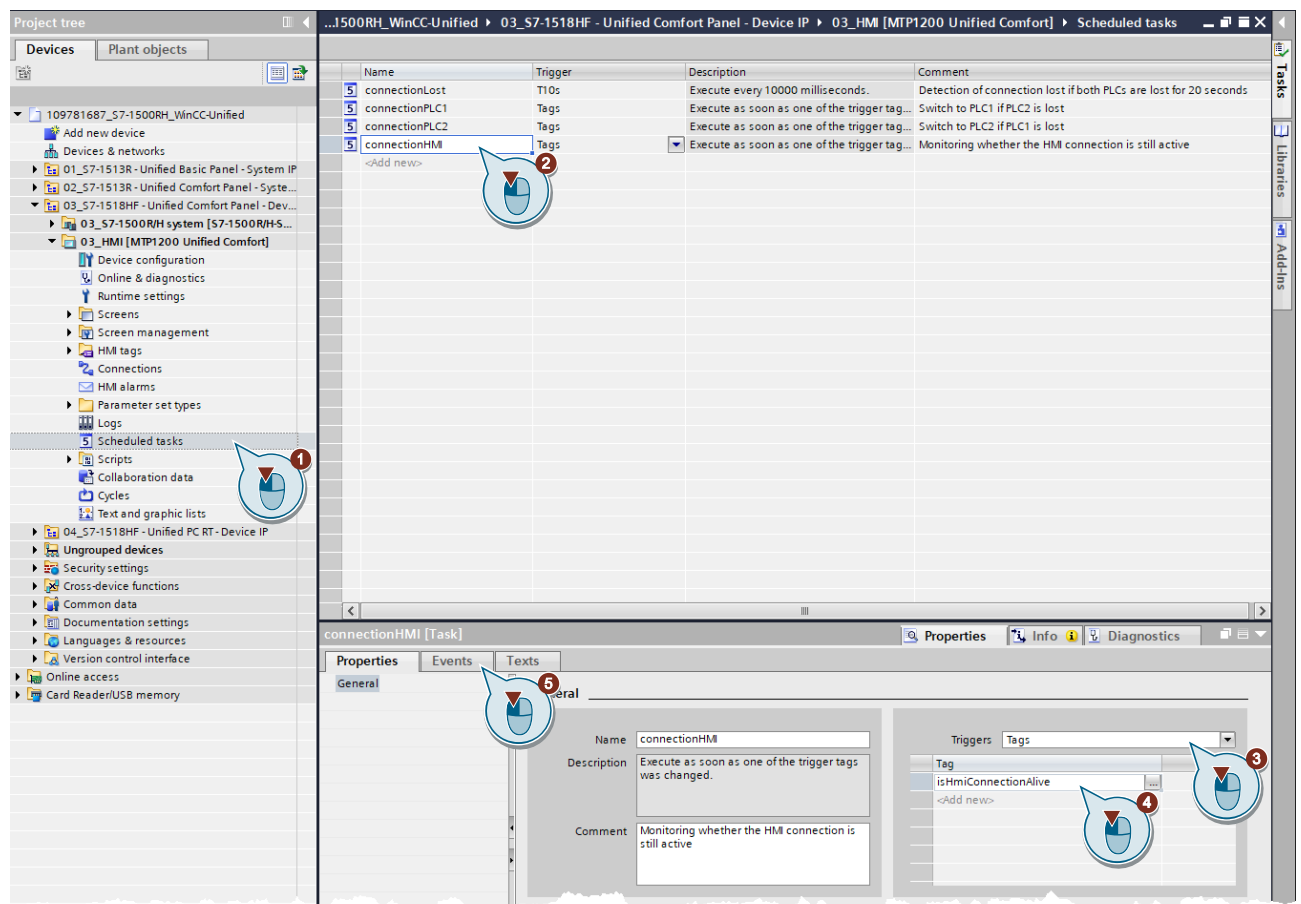
// Switch to PLC2 if PLC1 is lost
if (ts[0].Value > ts[6].Value && ts[2].Value === "Connected with PLC1") {
    // plc1Counter, changePlcCounter, lastConnectedPlc
    HMIRuntime.Connections.SysFct.ChangeConnection(ts[5].Value, ts[4].Value);
    // hmiConnectionName, plc2IpAddress
    ts[2].Value = "Connected with PLC2";    // lastConnectedPlc
    ts[0].Value = 0;                       // plc1Counter
}
ts.Write();
```

3.6.3. Scheduled tasks - connectionHMI

To edit the scheduled tasks:

1. Open the Scheduled Tasks section in the project tree
2. Select the relevant scheduled task
3. Use tags as triggers
4. Use the variable introduced in [For the configuration "01_Config"](#), which contains Lifebit from the HMI-Connection
5. The code is located under Events

Figure 3-16



Once a valid connection is re-established, the HMI counter (hmiCounter) is reset to zero. The system then sets the isHmiConnected flag to true, indicating that communication with at least one PLC is active again.

```
let ts = Tags.CreateTagSet([
    "hmiCounter",          // ts[0]
    "isHmiConnected"      // ts[1]
]);

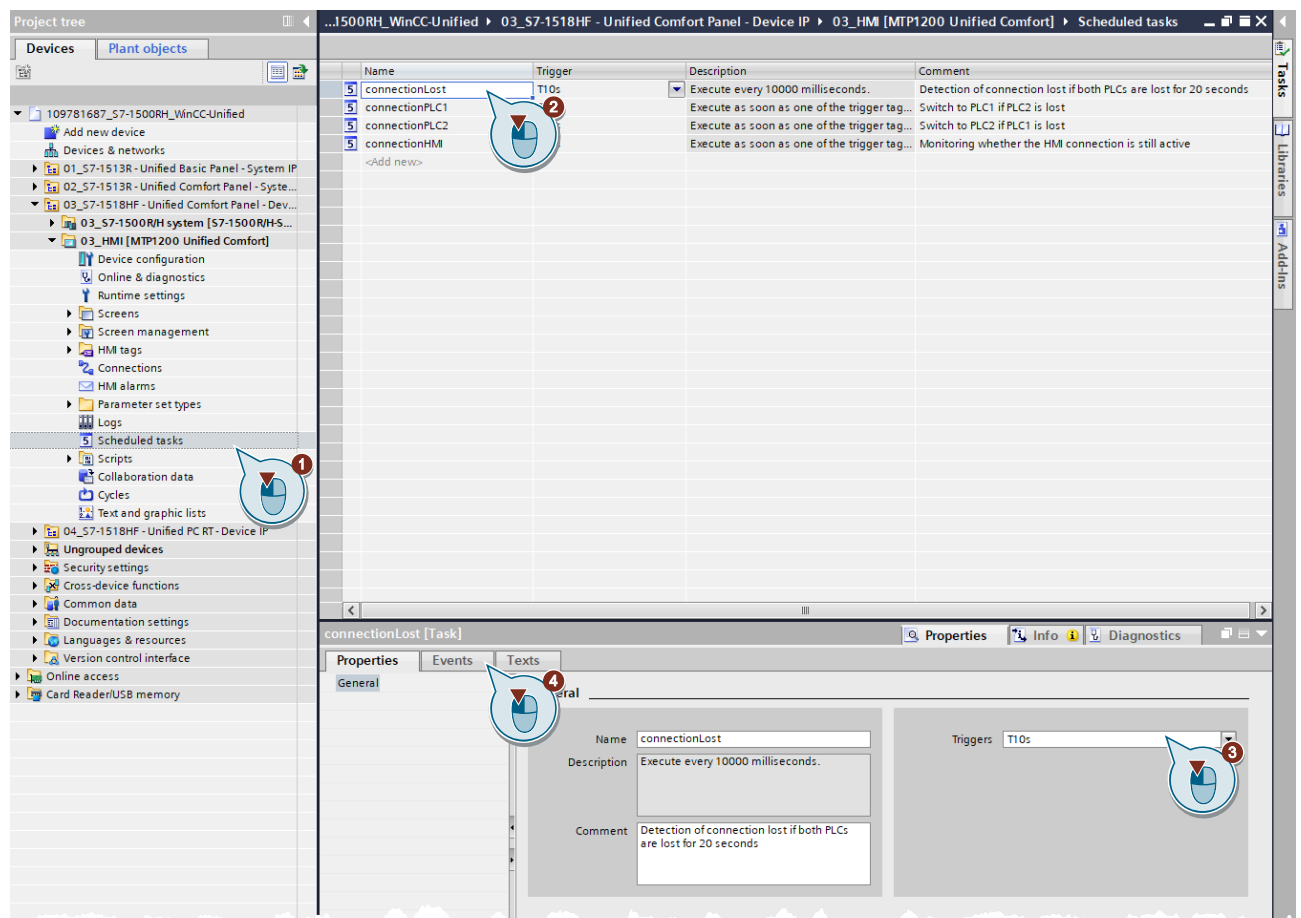
// Reset of the connection lost detection
ts[0].Value = 0;          // hmiCounter
ts[1].Value = true;      // isHmiConnected
ts.Write();
```

3.6.4. Scheduled tasks - connectionLost

To edit the scheduled tasks:

1. Open the Scheduled Tasks section in the project tree.
2. Select the relevant scheduled task.
3. The T10s trigger needs to be selected.
4. The code is located under Events.

Figure 3-17



An internal HMI counter (hmiCounter) is incremented continuously to monitor connection activity. If both PLCs remain unreachable for 20 seconds (equivalent to two cycles at 10-second intervals), the system sets the "isHmiConnected" flag to false.

Once the connection is marked as inactive, the system initiates a recovery procedure by alternating the communication between PLC1 and PLC2 in an attempt to re-establish a valid connection.

```
let ts = Tags.CreateTagSet([
    "plc1IpAddress",    // ts[0]
    "plc2IpAddress",    // ts[1]
    "hmiCounter",        // ts[2]
    "isHmiConnected",    // ts[3]
    "lastConnectedPlc",  // ts[4]
    "changePlcCounter",  // ts[5]
    "hmiConnectionName" // ts[6]
]);
ts.Read();

ts[2].Value = ts[2].Value + 1; // hmiCounter

//If both PLCs are unreachable for 20 seconds, an HMI connection lost is detected
if (ts[2].Value >= 2) {          // hmiCounter
    ts[3].Value = false;        // isHmiConnected
}

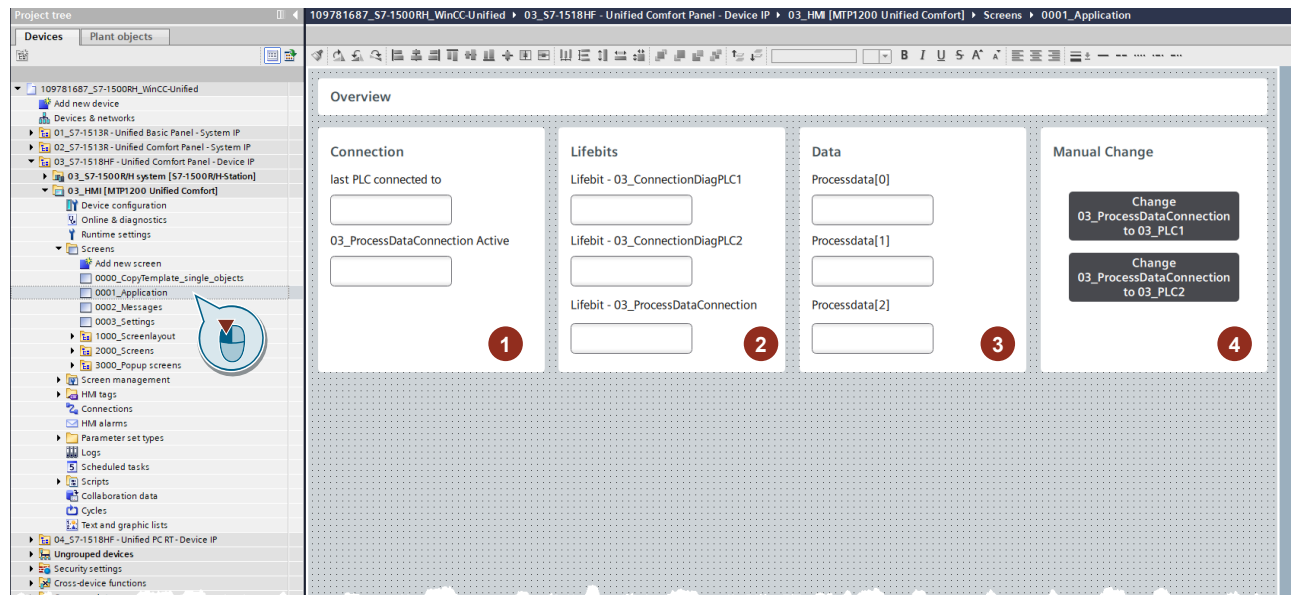
//After a HMI connection lost, connections are toggled to help restore a stable operational
state
if (ts[2].Value > 2 && ts[4].Value === "Connected with PLC1" ) {          // hmiCounter,
lastConnectedPlc
    HMIRuntime.Connections.SysFct.ChangeConnection(ts[6].Value, ts[1].Value); //
hmiConnectionName, plc2IpAddress
    HMIRuntime.Connections.SysFct.ChangeConnection(ts[6].Value, ts[0].Value); //
hmiConnectionName, plc1IpAddress
}

//After a HMI connection lost, connections are toggled to help restore a stable operational
state
if (ts[2].Value > 2 && ts[4].Value === "Connected with PLC2" ) {          // hmiCounter,
lastConnectedPlc
    HMIRuntime.Connections.SysFct.ChangeConnection(ts[6].Value, ts[0].Value); //
hmiConnectionName, plc1IpAddress
    HMIRuntime.Connections.SysFct.ChangeConnection(ts[6].Value, ts[1].Value); //
hmiConnectionName, plc2IpAddress
}
ts.Write();
```

3.7. HMI-Screen for monitoring

3.7.1. Screen for 03_S7-1518HF – Unified Comfort Panel – Device IP / 04_S7-1518HF – Unified PC RT – Device IP

Figure 3-18



Configurations "03_S7-1518HF – Unified Comfort Panel – Device IP" and "04_S7-1518HF – Unified PC RT – Device IP" display identical content across all screens. The monitor screen can be edited within screen **"0001_Application"**.

On the left side, under Connection (1), the interface shows which PLC was last selected as the active communication partner. Below that, a Boolean status field indicates whether the HMI connection used for process data exchange is currently active or inactive.

Another key element is the Lifebit monitoring for the three HMI connections (2). Each PLC has a dedicated Lifebit associated with its own HMI connection, along with an additional Lifebit for the connection used to transfer process data.

Process values are displayed in the next tab (3).

When using Device IP communication, the active communication partner can be switched manually (4).

NOTE

Manual Change is only possible after one Connection Switchover.

4. Appendix

4.1. Service and support

SiePortal

The integrated platform for product selection, purchasing and support - and connection of Industry Mall and Online support. The SiePortal home page replaces the previous home pages of the Industry Mall and the Online Support Portal (SIOS) and combines them.

- **Products & Services**
In Products & Services, you can find all our offerings as previously available in Mall Catalog.
- **Support**
In Support, you can find all information helpful for resolving technical issues with our products.
- **mySieportal**
mySiePortal collects all your personal data and processes, from your account to current orders, service requests and more. You can only see the full range of functions here after you have logged in.

You can access SiePortal via this address: sieportal.siemens.com

Technical Support

The Technical Support of Siemens Industry provides you fast and competent support regarding all technical queries with numerous tailor-made offers – ranging from basic support to individual support contracts.

Please send queries to Technical Support via Web form: support.industry.siemens.com/cs/my/src

SITRAIN – Digital Industry Academy

We support you with our globally available training courses for industry with practical experience, innovative learning methods and a concept that's tailored to the customer's specific needs.

For more information on our offered trainings and courses, as well as their locations and dates, refer to our web page: siemens.com/sitrain

Industry Online Support app

You will receive optimum support wherever you are with the "Industry Online Support" app. The app is available for iOS and Android:



4.2. Links and literature

Figure 4-1

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to this entry page of this application example https://support.industry.siemens.com/cs/ww/en/view/109781687
\3\	SIMATIC S7-1500 redundant system S7-1500R/H https://support.industry.siemens.com/cs/ww/en/view/109754833
\4\	HMI design with the HMI Template Suite https://support.industry.siemens.com/cs/af/en/view/91174767
\5\	SIMATIC WinCC Unified - Tips and Tricks for Scripting (JavaScript) https://support.industry.siemens.com/cs/af/en/view/109758536
\6\	SIMATIC WinCC Unified – Toolbox https://support.industry.siemens.com/cs/af/en/view/109770480

4.3. Change documentation

Figure 4-2

Version	Date	Modification
V1.0	10/2020	First version
V2.0	10/2025	Second version